

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

## 1<sup>st</sup> Semester of M. Pharm

### (Clinical Pharmacy & Pharmacology and Toxicology)

University Theory Examination March-April 2018

### PHCC004 ANALYTICAL TECHNIQUES

Date: 26.03.2018, Monday

Time: 10.00 a.m. to 01.00 p.m

Maximum Marks: 70

#### Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

#### SECTION-I

- Q.1**    **A)**    Discuss various factors affecting chemical shift. **05**  
          **B)**    Write a note on sampling techniques used in IR spectroscopy. **05**
- Q.2**    **A)**    Enlist the UV spectrophotometric methods for multicomponent analysis. Write a note on any one of them. **05**  
          **B)**    Enlist various ionization techniques used in Mass Spectroscopy. Write a note on MALDI. **05**

#### OR

- Q.2**    **A)**    Give advantages of UV spectrophotometric method over separation techniques. **05**  
          **B)**    Discuss instrumentation and applications of spectrofluorimeter. **05**
- Q.3**        **Answer the following (Any three)** **15**  
          **(A)**    Discuss Fragmentation rules in MS.  
          **(B)**    Give principle of Immunofluorescence Microscopy  
          **(C)**    Write a brief note on Cell imaging Techniques  
          **(D)**    Describe in brief FT-NMR.

#### SECTION-II

- Q.4**    **(A)**    Classify chromatographic methods. Discuss their mechanism of separation. **05**  
          **(B)**    Write applications of HPTLC method. **05**
- Q.5**    **(A)**    Write note on size exclusion chromatography. **05**  
          **(B)**    Discuss the theory and principles of electrophoresis. **05**
- Q.5**        **OR**  
          Discuss the Stationary phases and detectors used in GC. **10**
- Q.6**        **Answer the following (Any three)** **15**  
          **(A)**    Write note on ELISA.  
          **(B)**    Discuss different techniques for cell function assay.  
          **(C)**    Write note on applications of chromatographic methods  
          **(D)**    Write note on capillary electrophoresis.

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